

Generic Discriminants of Polynomial Tangent Pencils

Roy van Rijn
Independent researcher

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Abstract

Let H be a polynomial of degree $n \geq 3$ over an algebraically closed field of characteristic zero. The repeated-root locus of the pencil

$$H(W) - sW + t$$

is parameterized by

$$r \mapsto (H'(r), rH'(r) - H(r)).$$

We prove that, for H in a nonempty Zariski-open set, its projective closure is a rational plane curve of degree n , smooth at its unique point at infinity, with exactly $n - 2$ ordinary cusps and $\frac{1}{2}(n - 2)(n - 3)$ ordinary nodes, and no other singularities. The proof uses the contact of the graph of H with affine lines to exclude every nongeneric singularity before the genus formula is applied.

1 The theorem

Let k be an algebraically closed field of characteristic zero and write

$$H(W) = \sum_{j=2}^n h_j W^j, \quad h_n \neq 0. \quad (1.1)$$

Discarding the affine-linear part of H only changes coordinates in the (s, t) -plane, so (1.1) entails no loss of generality. Put

$$E_{s,t}(W) = H(W) - sW + t.$$

A repeated root r satisfies

$$s = H'(r), \quad t = rH'(r) - H(r). \quad (1.2)$$

Thus the reduced discriminant curve is the image of

$$\nu_H : \mathbb{A}^1 \longrightarrow \mathbb{A}^2, \quad r \mapsto (H'(r), rH'(r) - H(r)). \quad (1.3)$$

Geometrically, $(s, -t)$ records the affine line of slope s tangent to the graph $Y = H(W)$ at r . Cusps arise from flexes of the graph and nodes from lines tangent at two distinct points. This is a special one-parameter instance of the classical geometry of dual plane curves; see, for example, [1].

Theorem 1.1. *For every $n \geq 3$, there is a nonempty Zariski-open subset of the degree- n polynomials such that the projective discriminant curve of $H(W) - sW + t$ has:*

1. degree n and normalization $\mathbb{P}^1$;

2. one point at infinity, which is smooth;
3. exactly $n - 2$ ordinary cusps;
4. exactly $\frac{1}{2}(n - 2)(n - 3)$ ordinary nodes;
5. no other singularities.

The result concerns the reduced discriminant. No assertion is made here about the multiplicity with which a resultant defines it.

2 Normalization and infinity

Homogenizing (1.3) gives

$$\bar{\nu}_H : \mathbb{P}^1 \longrightarrow \mathbb{P}^2, \quad [R : U] \longmapsto [S_H : T_H : Z], \quad (2.1)$$

where

$$\begin{aligned} S_H &= \sum_{j=2}^n j h_j R^{j-1} U^{n-j+1}, \\ T_H &= \sum_{j=2}^n (j-1) h_j R^j U^{n-j}, \\ Z &= U^n. \end{aligned}$$

These forms have degree n and no common zero.

Proposition 2.1. *For every polynomial H of exact degree n , the map $\bar{\nu}_H$ is finite and birational onto an integral plane curve of degree n . Its image has a unique point at infinity, and that point is smooth.*

Proof. On the affine chart,

$$\frac{ds}{dr} = H''(r), \quad \frac{dt}{dr} = r H''(r).$$

At the generic point of the image this recovers $r = dt/ds$, so $\bar{\nu}_H$ is birational. A nonconstant morphism from a proper curve to its image is finite. Since (2.1) is given by base-point-free forms of degree n , birationality makes the image a curve of degree n .

The equation $Z = 0$ has the unique preimage $[1 : 0]$, whose image is $[0 : 1 : 0]$. Put $q = U/R$ and work in the target chart $T_H \neq 0$. Then

$$\frac{S_H}{T_H} = \frac{n}{n-1} q + O(q^2), \quad \frac{Z}{T_H} = \frac{q^n}{(n-1)h_n} + O(q^{n+1}). \quad (2.2)$$

The first series has nonzero linear coefficient. It is therefore a local parameter on the image, which is formally the graph of the second series as a function of the first. The point at infinity is smooth. $\square$

3 Contact strata

Let ℓ_r be the tangent line to the graph of H at r , and put

$$c(r) = \text{ord}_{W=r}(H(W) - \ell_r(W)).$$

One always has $c(r) \geq 2$. Differentiating (1.3) gives

$$\nu'_H(r) = H''(r)(1, r). \quad (3.1)$$

Thus $c(r) = 2$ gives a smooth immersed branch. If $c(r) = 3$, then $H'''(r) \neq 0$, and

$$\det(\nu''_H(r), \nu'''_H(r)) = 2H'''(r)^2 \neq 0. \quad (3.2)$$

The image branch is therefore an ordinary cusp. Contact of order at least four is the first bad one-branch event.

If two distinct parameters r and u have the same image and both branches are unramified, their tangent directions are $(1, r)$ and $(1, u)$. Their determinant is $u - r \neq 0$, so the two branches meet in an ordinary node.

It remains to show that every other event is nongeneric. For a contact pattern $\mu = (\mu_1, \dots, \mu_k)$, write

$$M_\mu(W) = \prod_{i=1}^k (W - r_i)^{\mu_i}, \quad m = \sum_i \mu_i.$$

An affine line having the prescribed contacts with the graph of H gives a factorization

$$H(W) - \ell(W) = M_\mu(W)Q(W), \quad \deg Q \leq n - m. \quad (3.3)$$

After affine-linear terms are discarded, such polynomials vary in a family of dimension at most

$$k + (n - m + 1). \quad (3.4)$$

The coefficient space in (1.1) has dimension $n - 1$.

Lemma 3.1. *Each of the contact patterns*

$$(4), \quad (3, 2), \quad (2, 2, 2) \quad (3.5)$$

is contained in a proper closed subset of the degree- n coefficient space.

Proof. For each pattern in (3.5), the dimension in (3.4) is $n - 2$, whenever the pattern is possible. The closure of the image of the factorization family (3.3) therefore has dimension at most $n - 2$, hence is proper in the $(n - 1)$ -dimensional coefficient space. Taking the closure also includes collisions among the marked contact points and degree drops in Q . $\square$

Lemma 3.2. *Outside the three closed loci in Lemma 3.1, every affine singularity of the discriminant is an ordinary cusp or an ordinary node.*

Proof. A nonordinary single branch has contact at least four and belongs to the (4)-locus. If an image point has two normalization preimages and one branch is ramified, its contact pattern dominates (3, 2). If it has at least three preimages, any three give the pattern (2, 2, 2). The only remaining singular image has exactly two unramified preimages, and (3.1) makes their tangent directions distinct. $\square$

Proposition 2.1 shows uniformly that no additional singularity occurs at infinity. The complement of the three bad loci is therefore a nonempty Zariski-open set on which the discriminant has only ordinary cusps and nodes.

4 The count

On the open set just constructed, H'' is squarefree. Its $n - 2$ roots are precisely the ramification points of (1.3), and (3.2) shows that their images are $n - 2$ distinct ordinary cusps.

The discriminant is an integral plane curve of degree n with normalization $\mathbb{P}^1$. Its arithmetic genus is

$$p_a = \frac{(n-1)(n-2)}{2},$$

whereas its geometric genus is zero. Ordinary nodes and ordinary cusps both have delta invariant one. If N is the number of nodes, the genus formula gives

$$\frac{(n-1)(n-2)}{2} = (n-2) + N.$$

Hence

$$N = \frac{(n-2)(n-3)}{2},$$

which proves Theorem 1.1.

Acknowledgments and declaration of generative AI use

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References

- [1] A. Yuran, *Plücker formulas for plane algebraic curves with a given Newton polygon*, arXiv:2204.04626.